AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 30 · TEACHER GUIDE

Dataset shift in neural models

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make every percentage traceable to counts.

Use the mixture graphic and checked examples to separate data monitoring from measured accuracy.

Keep one habit: Model pooled and weighted routes → diagnose denominators → assess fresh P/Q counts → plan independent follow-up.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain how a changed **input** mix can change overall accuracy

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

No. Confidence is a model **output**, not a guarantee of familiarity or correctness. Check **input** conditions and independently verified outcomes rather than treating confidence as proof.

Opening retrieval

Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a predicted class, confidence score and observed correctness

Reference example, not the only acceptable wording: Teacher explanation: B predicts 1 with confidence 0.90 and is wrong. E predicts 0 with confidence 0.55 and is correct. F predicts 0 with confidence 0.90 and is wrong.

F's low score for class 1 corresponds to a high score for class 0, the class selected.

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate vector length and normalise a nonzero vector

Reference example, not the only acceptable wording: Teacher explanation: Lengths are 5, 10 and 5. Unit vectors are [0.6, 0.8], [0.6, 0.8] and [-0.6, -0.8].

The length difference between Q and A disappears; the opposite direction of C is preserved.

Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

Look for: Check whether a transformation preserves the task label

Reference example, not the only acceptable wording: Teacher explanation: Original d = -2 and class 0. Row swapping leaves d = -2 and class 0. Column swapping gives rows [1, 0], [1, 0], d = 2 and class 1.

Pictured end goal

By the end, I can explain **dataset shift** in neural models using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Uncertainty and failure analysis

Similarity and normalisation

Augmentation and leakage

For the compact alternative schedule, use two 45-minute sessions. Work from counts before percentages and keep the frozen model separate from the changing sample mix.

The exact equal group rates are constructed for arithmetic; they do not establish a population law. A **monitoring** alert should lead to **evidence** gathering rather than an automatic causal claim or re**training** decision.

Outcomes

Explain how a changed **input** mix can change overall accuracy

Calculate group and overall results with the correct denominators

Plan a shift investigation while preserving independent evaluation

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach weighted counts before interpreting a changing score

> Keep the main 90% clear rate separate from the picture’s 80% bright rate.

Definitions, controlled scope and complete main inputs

A neural image classifier developed using mostly clear pictures may later receive mostly dim pictures. Its learned weights can stay unchanged while its overall results change.

Dataset shift concerns changes in the data encountered, including **input**s, label frequencies or the relationship between **input**s and labels.

A shift does not always reduce accuracy, and an accuracy change alone does not identify its cause.

Use the fictional audit below. Clear and dim are disjoint groups defined before the audit, and every case belongs to exactly one group. The same model checkpoint, pre**process**ing and **prediction** threshold are used throughout.

Counts are constructed teaching data, not results from a deployed model. Both cohorts have checked reference labels.

Cohort	Group	Cases	Correct
Development audit	Clear	80	72
Development audit	Dim	20	10
New-use audit	Clear	20	18
New-use audit	Dim	80	40

Checked main arithmetic and visual counts

Clear accuracy is 72 / 80 = 90 percent in development and 18 / 20 = 90 percent in new use. Dim accuracy is 10 / 20 = 50 percent and 40 / 80 = 50 percent.

The observed group rates are unchanged, but the group proportions are different.

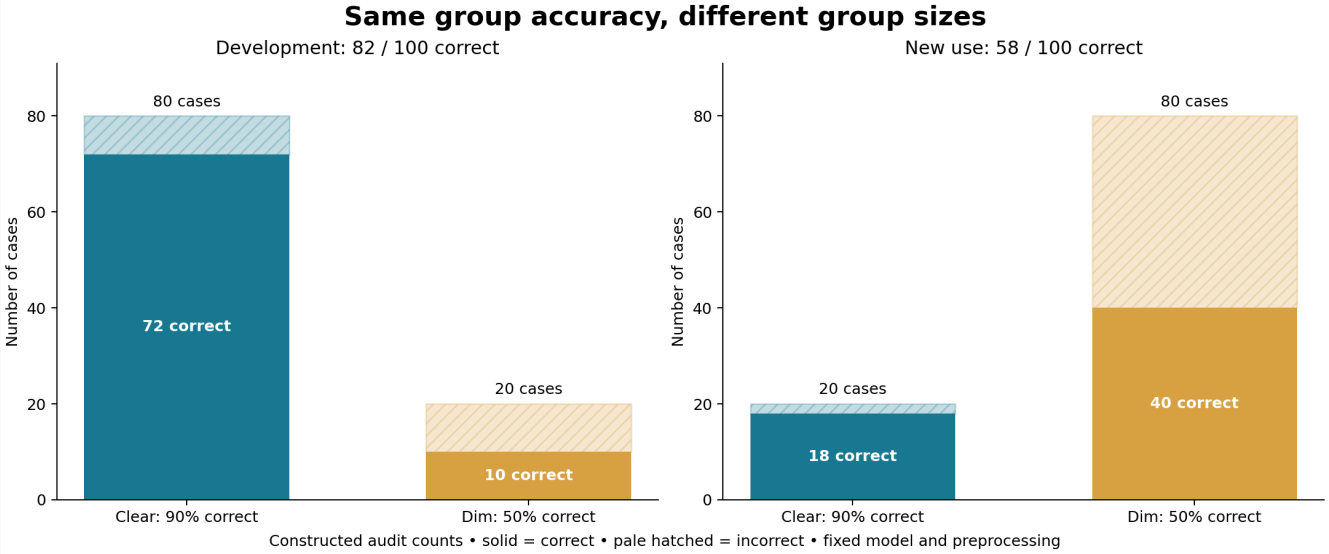
Development accuracy is (72 + 10) / (80 + 20) = 82 percent. New-use accuracy is (18 + 40) / (20 + 80) = 58 percent. The drop is 24 percentage points.

Taking the unweighted mean of 90 and 50 gives 70 percent, which is not either cohort’s overall accuracy because neither cohort has equal group sizes.

Each pair is (cases, correct), and the function assumes integer counts for non-overlapping groups. It rejects empty **input**s, groups with no cases and impossible correct counts.

A group with no observed cases has an undefined empirical accuracy, not zero accuracy. Do not count the same example in two groups when forming an overall total.

Equivalently, weight each group accuracy by its share of cases: development is 0.80 times 0.90 + 0.20 times 0.50 = 0.82; new use is 0.20 times 0.90 + 0.80 times 0.50 = 0.58.



Main equal-mixture key and evidence limits

The same group rates meet a different mix

GROUP	DEVELOPMENT SHARE	NEW-USE SHARE
Clear, 90% correct	80 of 100	20 of 100
Dim, 50% correct	20 of 100	80 of 100

The weaker group becomes more common. Overall mistakes rise from 18 of 100 to 42 of 100 without any change in these observed within-group accuracy rates.

Guided: 45 clear plus 25 dim cases are correct, giving 70 of 100 or 70 percent. This assumes exactly the stated group rates hold in that cohort. Exit: the mix change is observed, but accuracy is unknown without checked labels.

Gather a labelled audit and check model/pipeline versions. Cases used to choose or fit a change cannot also provide independent final **evidence**.

Reserve new appropriate cases and compare original and revised models on the same cases with group and overall results.

The arithmetic explains this constructed aggregate difference. It does not prove that dim lighting caused each real error or that these group rates will stay fixed in future samples.

Report group counts and uncertainty about generalisation. Review specific errors and reference labels as in Week 29.

> **If a new cohort has 50 clear and 50 dim cases, and exactly 90 percent and 50 percent are correct respectively, how many correct cases and what overall accuracy result? State the assumption behind this calculation.**

Evidence-to-action checks

Evidence or check	Permitted conclusion / next action
More dim inputs, no new checked labels	Changed input mix observed; current accuracy unavailable
Same input histogram, changed label policy	Relationship can change without an obvious input-summary change
Different resize, channel order or units	Check implementation versions before attributing results to population shift
New paired lighting comparisons	Keep scene identity and label fixed; keep related source images in one split
Retraining uses these new errors	Those cases become development evidence; reserve new appropriate final cases
Original versus revised model comparison	Run both on identical reserved cases; retain baseline, subgroup counts and workload

Evidence or check	Permitted conclusion / next action
No cases in a declared group	Empirical group accuracy unavailable; do not record zero accuracy

Separate pictured inputs and scope

Separate within-setting performance from changing setting proportions.

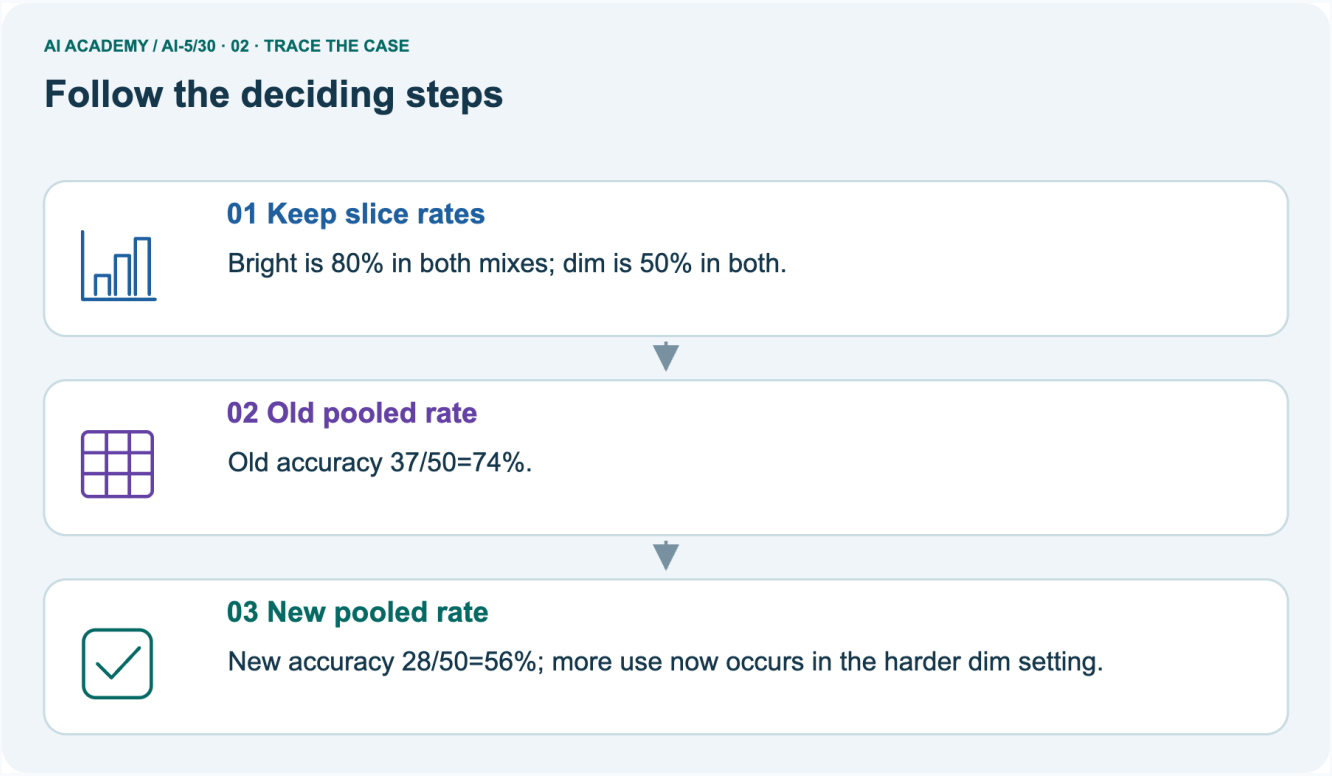
Toy old-use mix: bright 40 cases 32 correct; dim 10 cases 5 correct. New-use mix: bright 10 cases 8 correct; dim 40 cases 20 correct.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate 50-case example: bright accuracy is 80%, not the main clear-group rate of 90%. Old picture counts 40 bright/32 correct plus 10 dim/5 correct give 37/50 = 74%. New counts 10 bright/8 correct plus 40 dim/20 correct give 28/50 = 56%. Returning to the old proportions under the same rates returns 74%.

This isolates a mixture calculation; it does not establish future rates or causal lighting effects.

Picture trace and restored-mixture key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep slice rates	Bright is 80% in both mixes; dim is 50% in both.
2. Old pooled rate	Old accuracy 37/50=74%.
3. New pooled rate	New accuracy 28/50=56%; more use now occurs in the harder dim setting.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A pooled rate depends on how often each condition appears as well as performance within that condition. Bright and dim accuracy stay fixed in this example. The new use contains more dim cases, so its pooled accuracy falls.

Report both group counts and group rates before concluding that the network itself deteriorated.

Real shift can also change labels, sensors and within-slice behaviour; this toy decomposition isolates only mixture change.

Change: New-use mix returns to40 bright and 10 dim with the same slice rates.

Prediction: Pooled accuracy returns to74%.

Reason: The mixture weights change, while each slice’s conditional performance stays fixed.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Dataset shift	A change in the distribution of inputs, labels or their relationship between settings
Subgroup	A declared subset of examples sharing a relevant condition
Group mix	The proportions of examples belonging to different groups
Monitoring	Repeatedly checking recorded inputs, outputs and available outcomes against a reference

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

The same group rates meet a different mix

GROUP	DEVELOPMENT SHARE	NEW-USE SHARE
Clear, 90% correct	80 of 100	20 of 100
Dim, 50% correct	20 of 100	80 of 100

The weaker group becomes more common. Overall mistakes rise from 18 of 100 to 42 of 100 without any change in these observed within-group accuracy rates.

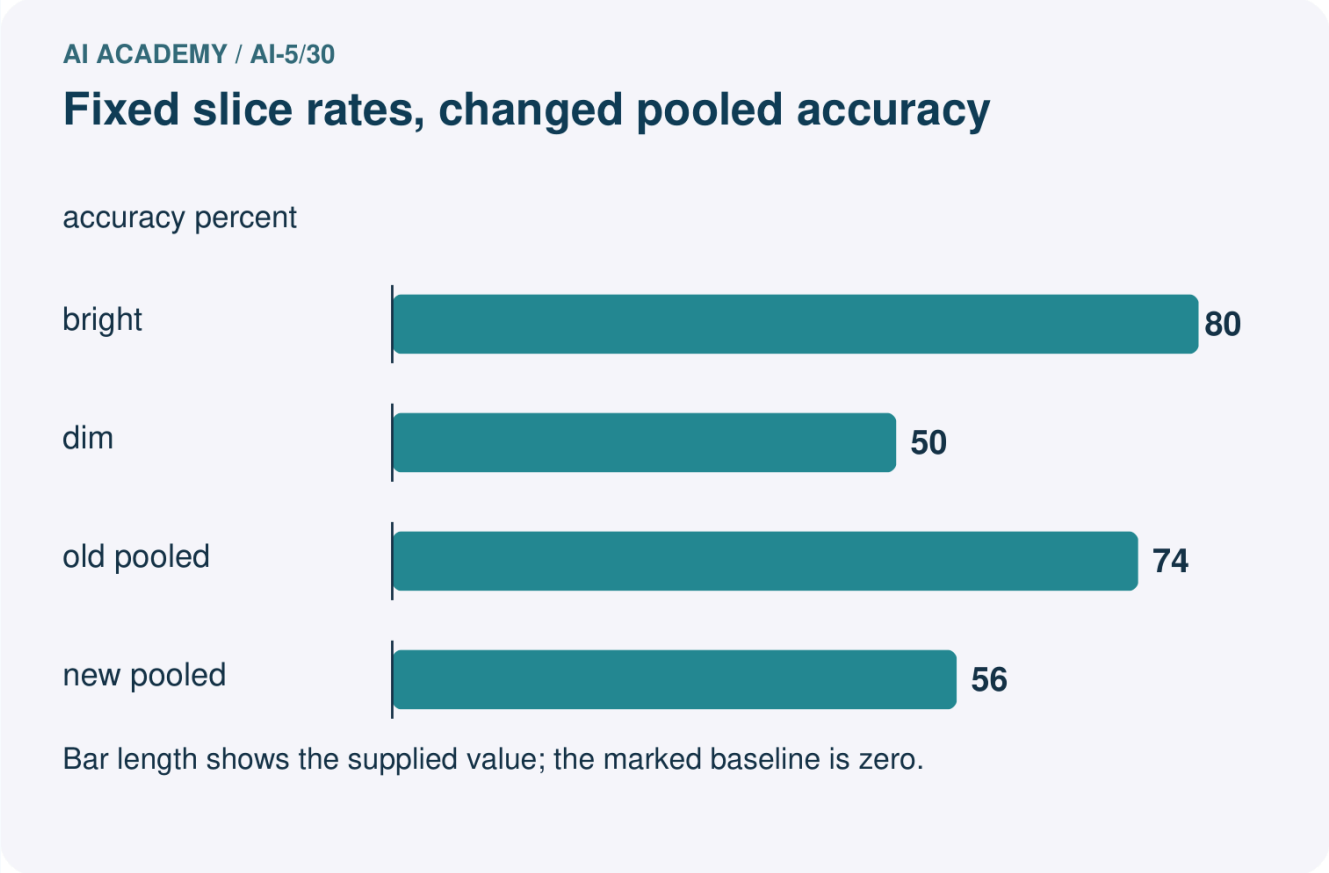
Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Dataset shift in neural models

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/30

Dataset shift in neural models

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep slice rates

Bright is 80% in both mixes; dim is 50% in both.



2. Old pooled rate

Old accuracy 37/50=74%.



3. New pooled rate

New accuracy 28/50=56%; more use now occurs in the harder dim setting.

CHANGE ONE THING

New-use mix returns to40 bright and 10 dim with the same slice rates.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Calculate overall correct counts, accuracy and errors in both cohorts, plus the percentage-point drop. Explain why averaging 90 percent and 50 percent without weights gives the wrong overall result here.

Teacher check: Teacher explanation: Development has 82 correct and 18 errors out of 100, or 82 percent accuracy. New use has 58 correct and 42 errors out of 100, or 58 percent. The drop is 24 percentage points.

An unweighted mean gives 70 percent and ignores the unequal group sizes.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Dataset shift** in neural models. Proposed explanation: “Average the percentages regardless of group sizes”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Overall accuracy is total correct divided by total cases; group accuracies need weights proportional to group size.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Did the supplied model’s dim-light accuracy deteriorate? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

No, it remains 50%. The pooled drop from 74% to56% reflects a larger proportion of dim cases, not changed within-slice rates. Changed-case check: Pooled accuracy returns to74%.

The mixture weights change, while each slice’s conditional performance stays fixed.

Changed-case reasoning: Pooled accuracy returns to74%. The mixture weights change, while each slice’s conditional performance stays fixed.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide separate columns for cases, correct, errors and group share. Use 100 counters or a drawn 100-cell grid to change the mix while keeping group rates fixed.

Extension: with clear share s and fixed rates 0.90 and 0.50, overall accuracy is 0.50 + 0.40*s. Test s = 0, 0.5 and 1, and explain why observed data need not keep those rates fixed.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve Week 29's warning that confident outputs can be wrong on unfamiliar inputs.
7 to 18 minutes	Read the cohort table and complete W1. Mark fixed checkpoint and pipeline versus changed mix.
18 to 32 minutes	Compute totals and W2. Compare the weighted result with the tempting unweighted average.
32 to 45 minutes	Trace the optional count function and the visual; explain why group sizes affect overall accuracy.
Session 2 start to 8 minutes	Complete the guided balanced cohort and name the fixed-rate assumption.
8 to 20 minutes	Complete W3 and discuss missing labels, pipeline changes and possible learned shortcuts.
20 to 35 minutes	Collect W4 independently, requiring counts, group rates, both totals and percentage points.
35 to 45 minutes	Use W5 and the exit response to protect independent evaluation. Give feedback, then a fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main rates .90/.50, totals .82/.58, fall 24 percentage points; equal mix .70 under stated rates. No labels means no measured current accuracy. Independent P .80 and Q .50 in both cohorts; development 68/100, new use 56/100, fall 12 percentage points. Equal P/Q mix gives .65. Do not average group percentages without group-size weights. No cases means unavailable group rate.

Protect new final evidence from retraining and source-image overlap.

Diagnose the current misconceptions

Average the percentages regardless of group sizes: Overall accuracy is total correct divided by total cases; group accuracies need weights proportional to group size.

A shift alert proves a loss of accuracy: **Input** changes may help, hurt or leave accuracy unchanged; verified outcomes are needed to measure performance.

An unchanged **input** summary proves nothing changed: A summary can miss relevant differences, and the **input**-label relationship or labelling policy can change without an obvious **input** shift.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A separate audit has development groups P: 60 cases, 48 correct and Q: 40 cases, 20 correct. New-use groups are P: 20 cases, 16 correct and Q: 80 cases, 40 correct.

Compute each group rate, both overall accuracies, and the percentage-point change. Under the same group rates, what accuracy would equal-sized P and Q groups have?

Teacher explanation: P is 80 percent correct and Q is 50 percent correct in both cohorts. Development accuracy is 68 of 100, or 68 percent; new-use accuracy is 56 of 100, or 56 percent. The drop is 12 percentage points.

Equal-sized groups at those rates give 65 percent accuracy.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Audit the two cohorts

From the table, calculate clear and dim accuracy in each cohort and the clear/dim proportions. Which observed quantities change, and which remain the same?

Teacher explanation: Clear accuracy is 90 percent in both cohorts; dim accuracy is 50 percent in both. Development has 80 percent clear and 20 percent dim; new use has 20 percent clear and 80 percent dim.

The mix changes while observed group rates remain the same.

W2 Use the right denominator

Calculate overall correct counts, accuracy and errors in both cohorts, plus the percentage-point drop. Explain why averaging 90 percent and 50 percent without weights gives the wrong overall result here.

Teacher explanation: Development has 82 correct and 18 errors out of 100, or 82 percent accuracy. New use has 58 correct and 42 errors out of 100, or 58 percent. The drop is 24 percentage points.

An unweighted mean gives 70 percent and ignores the unequal group sizes.

W3 Separate an alert from an outcome

New-use **monitoring** shows more dim images but reference labels have not arrived. What can you report now, what can you not calculate, and what checks should precede blaming the neural network?

Teacher explanation: Report the observed change in group proportions with counts and period. Current accuracy and error patterns need verified outcomes.

Check checkpoint and pre**process**ing versions, group definitions, **input** quality and label policy; collect a representative labelled audit sample before concluding how performance changed.

W5 Protect the comparison

A team retrains using new-use errors, then evaluates on those same images and their brightened copies. Explain why the score is not independent **evidence** and propose a comparison that includes the original model.

Teacher explanation: The evaluated cases influenced **training**, and transformed copies share source content. Keep related images in one split and reserve untouched source groups or an appropriate later period.

Evaluate original and revised models on identical reserved examples, with checked labels, group counts and rates, overall accuracy, and documented pre**process**ing.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: 45 clear plus 25 dim cases are correct, giving 70 of 100 or 70 percent. This assumes exactly the stated group rates hold in that cohort. Exit: the mix change is observed, but accuracy is unknown without checked labels.

Gather a labelled audit and check model/pipeline versions. Cases used to choose or fit a change cannot also provide independent final **evidence**.

Reserve new appropriate cases and compare original and revised models on the same cases with group and overall results.

W1 Audit the two cohorts

W2 Use the right denominator

W3 Separate an alert from an outcome

W4 Analyse a new mixture independently

A separate audit has development groups P: 60 cases, 48 correct and Q: 40 cases, 20 correct. New-use groups are P: 20 cases, 16 correct and Q: 80 cases, 40 correct.

Compute each group rate, both overall accuracies, and the percentage-point change. Under the same group rates, what accuracy would equal-sized P and Q groups have?

Teacher explanation: P is 80 percent correct and Q is 50 percent correct in both cohorts. Development accuracy is 68 of 100, or 68 percent; new-use accuracy is 56 of 100, or 56 percent. The drop is 12 percentage points.

Equal-sized groups at those rates give 65 percent accuracy.

W5 Protect the comparison

Independent assessment and fresh retry

W4 requires group rates 80 and 50 percent, overall accuracies 68 and 56 percent, a 12-percentage-point drop, and equal-group accuracy 65 percent.

Fresh retry: development P has 75 cases with 60 correct; Q has 25 with 10 correct. New use P has 25 with 20 correct; Q has 75 with 30 correct. Group rates are 80 and 40 percent in both.

Overall accuracy changes from 70 to 50 percent, a 20-percentage-point drop. Equal-sized groups at those rates give 60 percent.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **dataset shift** in neural models **evidence** sheet to Image-pattern Network Investigation. Show how you can explain how a changed **input** mix can change overall accuracy.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide separate columns for cases, correct, errors and group share. Use 100 counters or a drawn 100-cell grid to change the mix while keeping group rates fixed.

Extension: with clear share s and fixed rates 0.90 and 0.50, overall accuracy is 0.50 + 0.40*s. Test s = 0, 0.5 and 1, and explain why observed data need not keep those rates fixed.

Week 31 measures efficiency and resource use while keeping task quality, hardware and comparison conditions explicit.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/31, ai-5/33, ai-5/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain how a changed **input** mix can change overall accuracy

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Calculate overall correct counts, accuracy and errors in both cohorts, plus the percentage-point drop.

Explain why averaging 90 percent and 50 percent without weights gives the wrong overall result here.

Reasoned practice answer: Teacher explanation: Development has 82 correct and 18 errors out of 100, or 82 percent accuracy. New use has 58 correct and 42 errors out of 100, or 58 percent. The drop is 24 percentage points.

An unweighted mean gives 70 percent and ignores the unequal group sizes.

Check every learner before discussion: Use the worked case for **Dataset shift** in neural models. Proposed explanation: “Average the percentages regardless of group sizes”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Overall accuracy is total correct divided by total cases; group accuracies need weights proportional to group size.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: New-use mix returns to40 bright and 10 dim with the same slice rates. Require a **prediction** and a reason before comparing. Expected result: Pooled accuracy returns to74%.

Reason: The mixture weights change, while each slice's conditional performance stays fixed.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

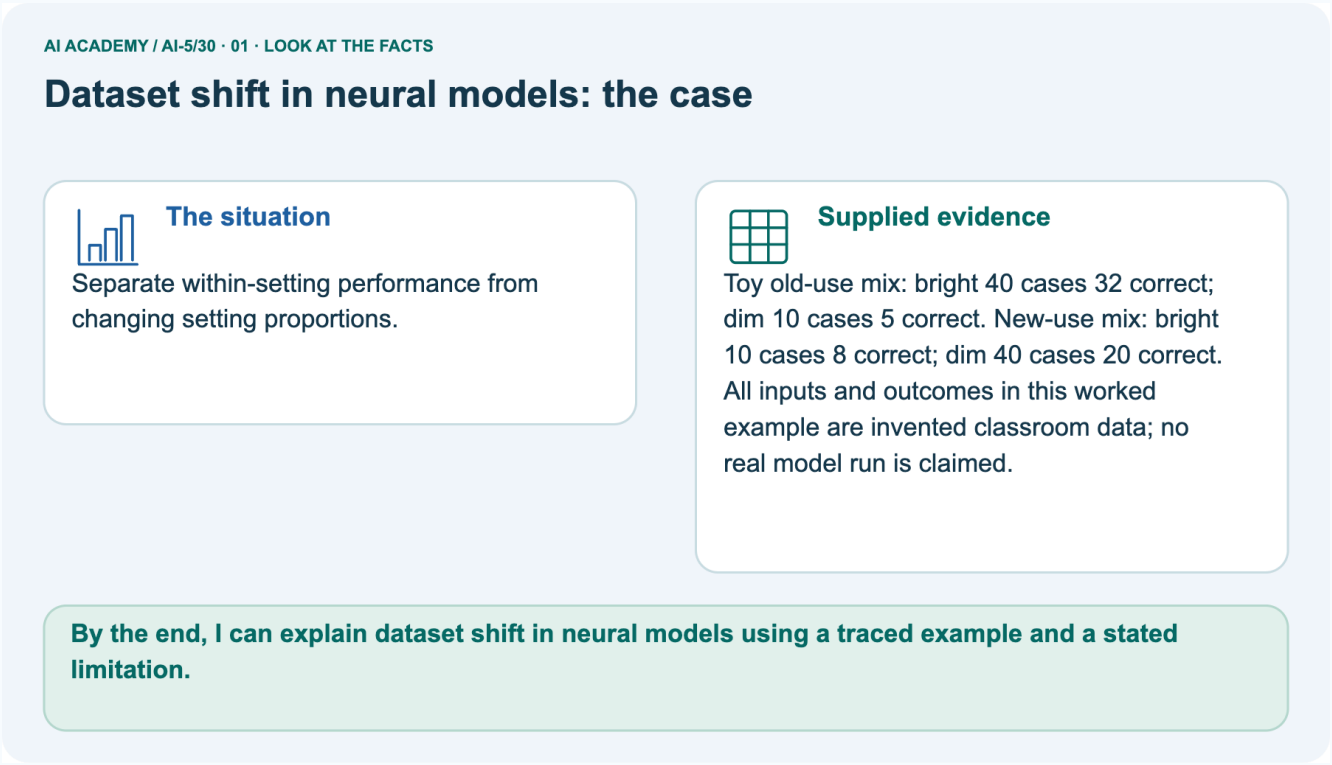
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Dataset shift in neural models: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

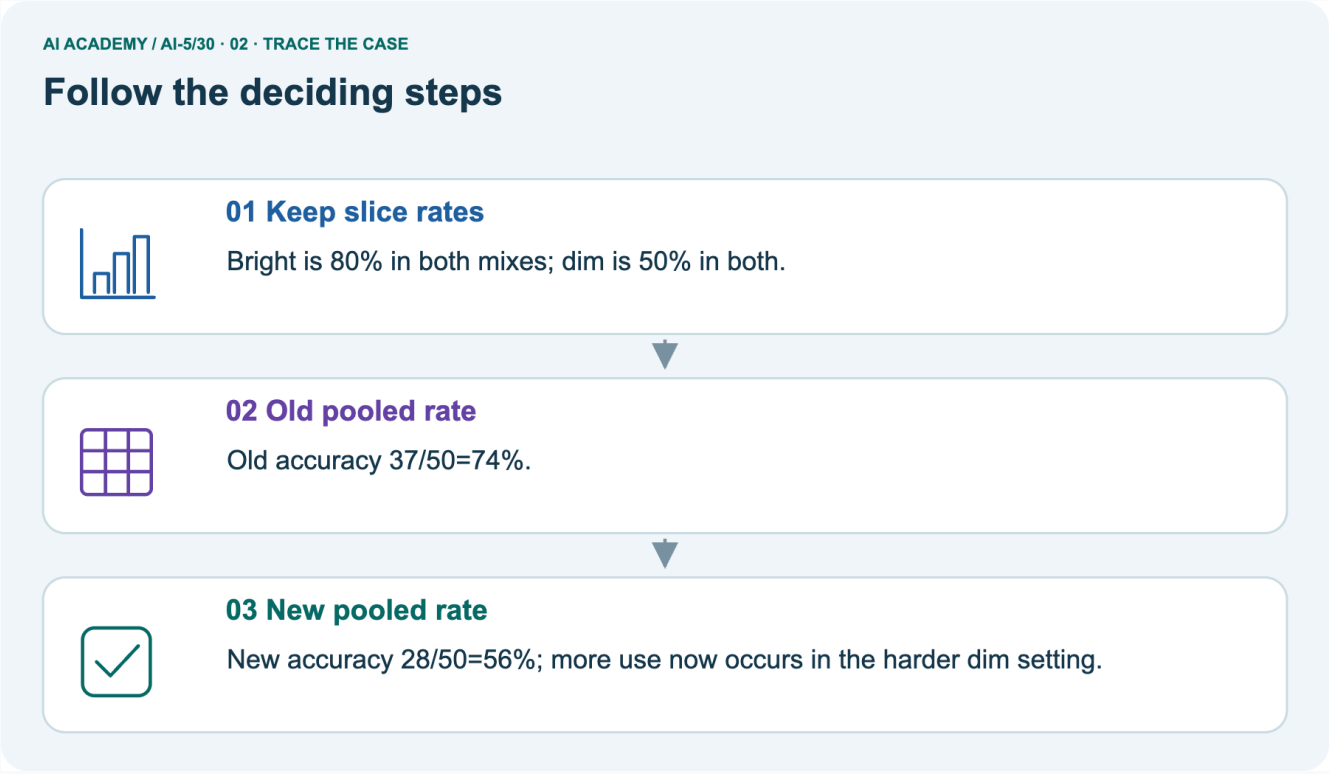
Separate within-setting performance from changing setting proportions. Supplied facts: Toy old-use mix: bright 40 cases 32 correct; dim 10 cases 5 correct. New-use mix: bright 10 cases 8 correct; dim 40 cases 20 correct.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



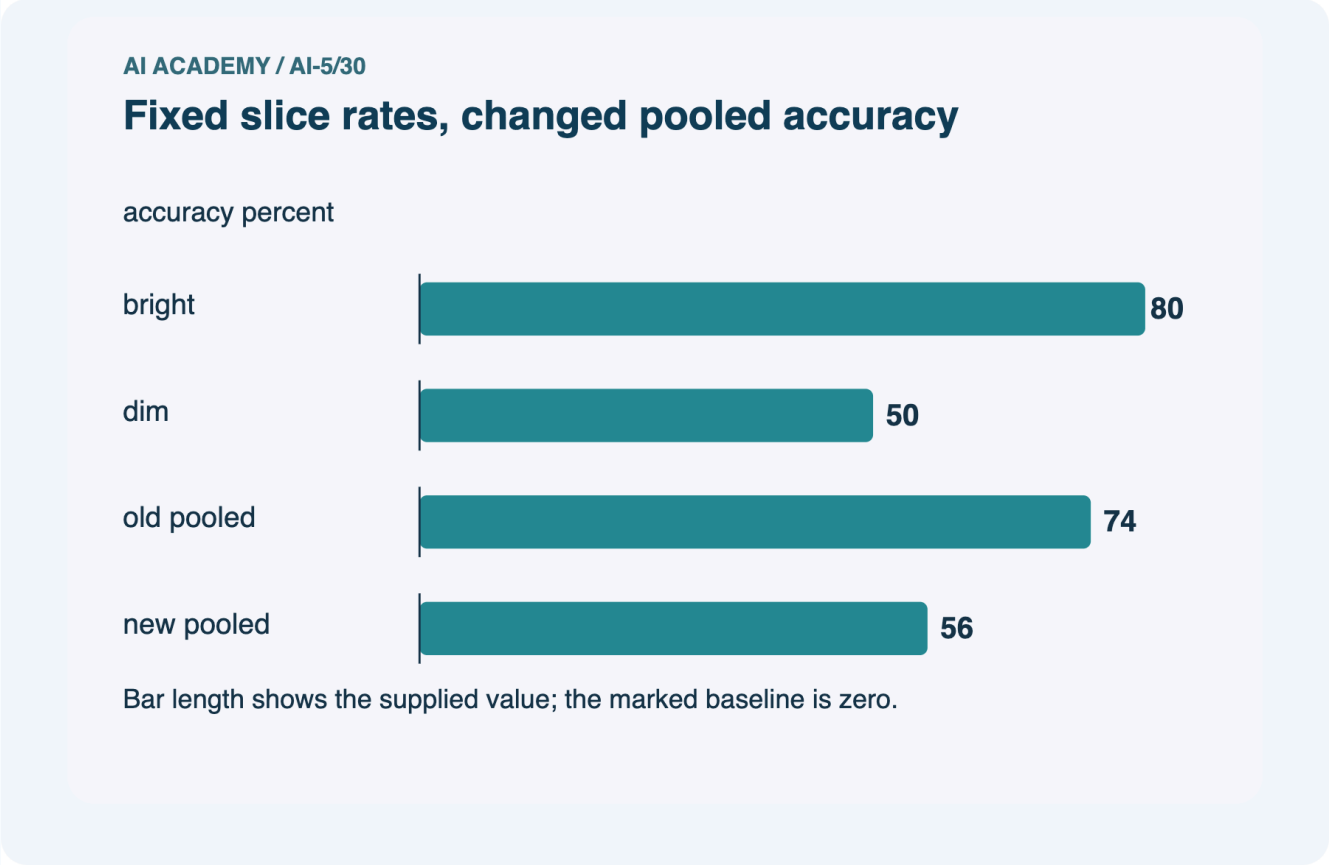
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep slice rates: Bright is **80%** in both mixes; dim is 50% in both. → Old pooled rate: Old accuracy 37/50=74%. → New pooled rate: New accuracy 28/50=56%; more use now occurs in the harder dim setting.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Fixed slice rates, changed pooled accuracy



Follow the labelled picture with the matching explanation. Open the vector diagram

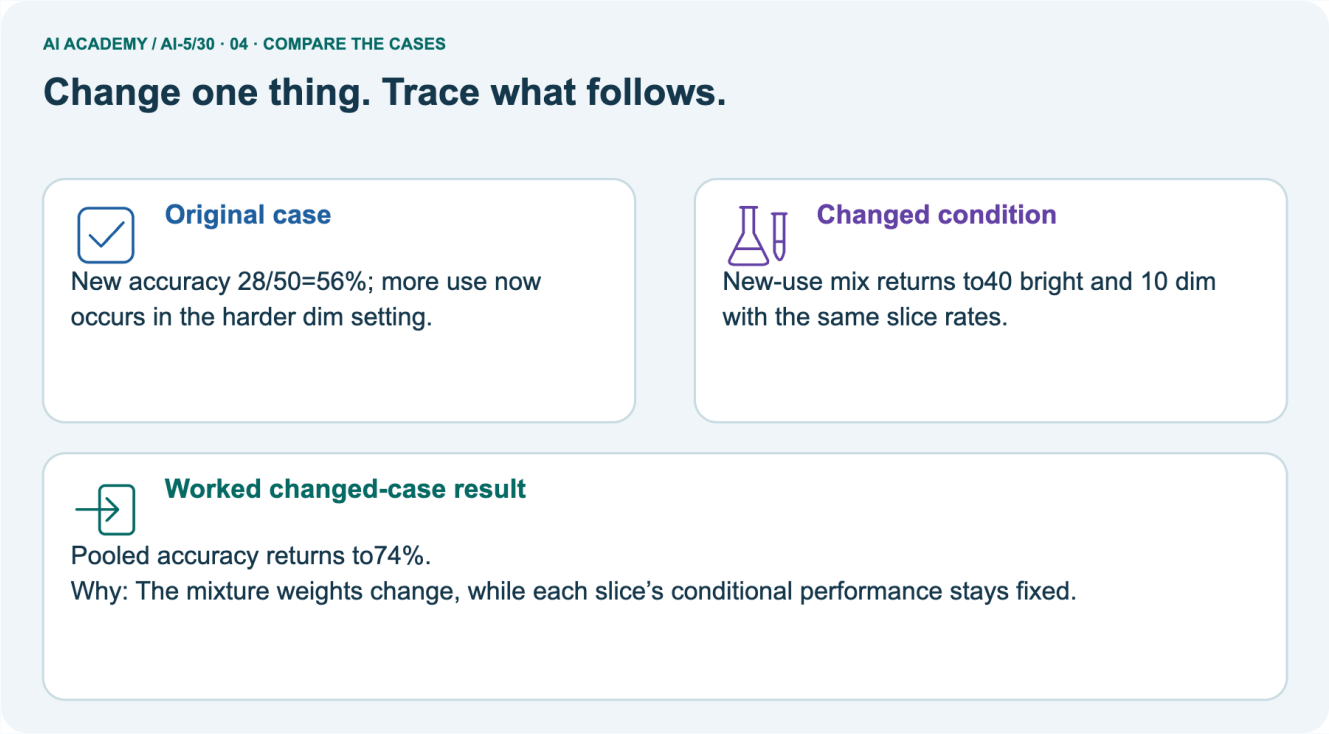
Fixed slice rates, changed pooled accuracy. Units: accuracy percent. bright: 80. dim: 50. old pooled: 74. new pooled: 56.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: New accuracy 28/50=56%; more use now occurs in the harder dim setting. Changed condition: New-use mix returns to40 bright and 10 dim with the same slice rates. Expected result: Pooled accuracy returns to74%.

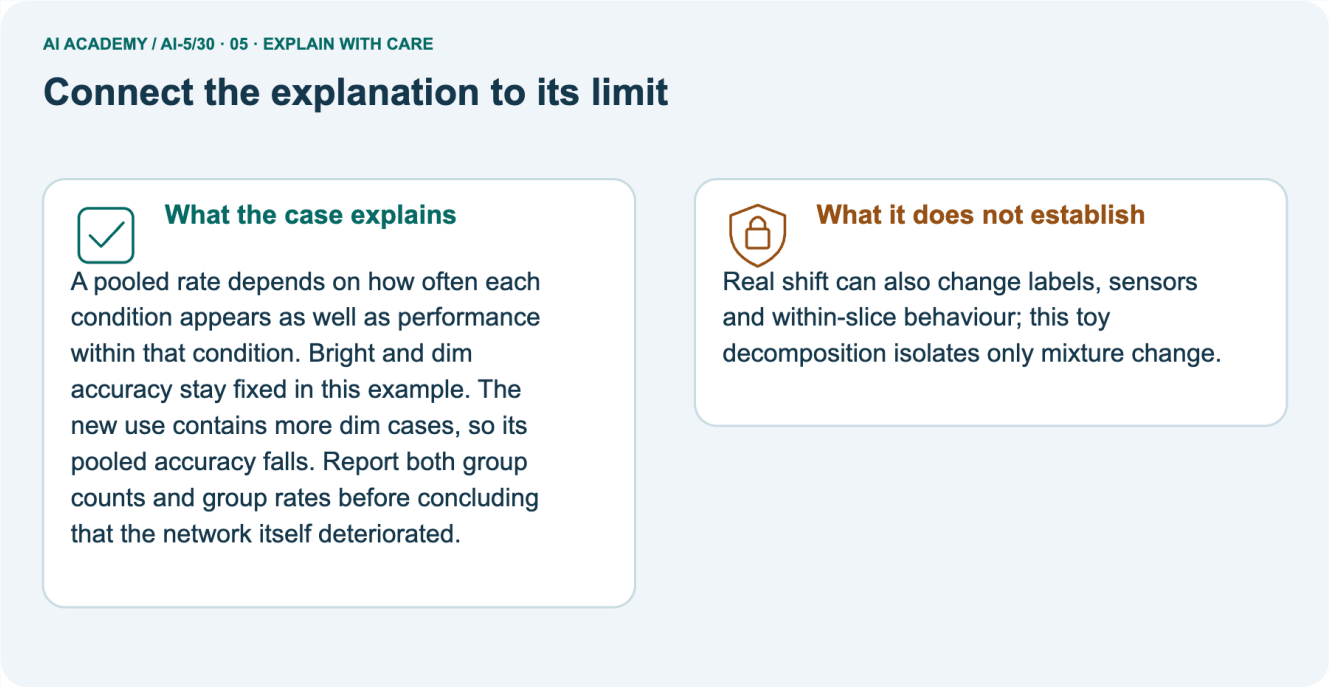
Reason: The mixture weights change, while each slice's conditional performance stays fixed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The mixture weights change, while each slice's conditional performance stays fixed.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A pooled rate depends on how often each condition appears as well as performance within that condition. Bright and dim accuracy stay fixed in this example. The new use contains more dim cases, so its pooled accuracy falls.

Report both group counts and group rates before concluding that the network itself deteriorated.

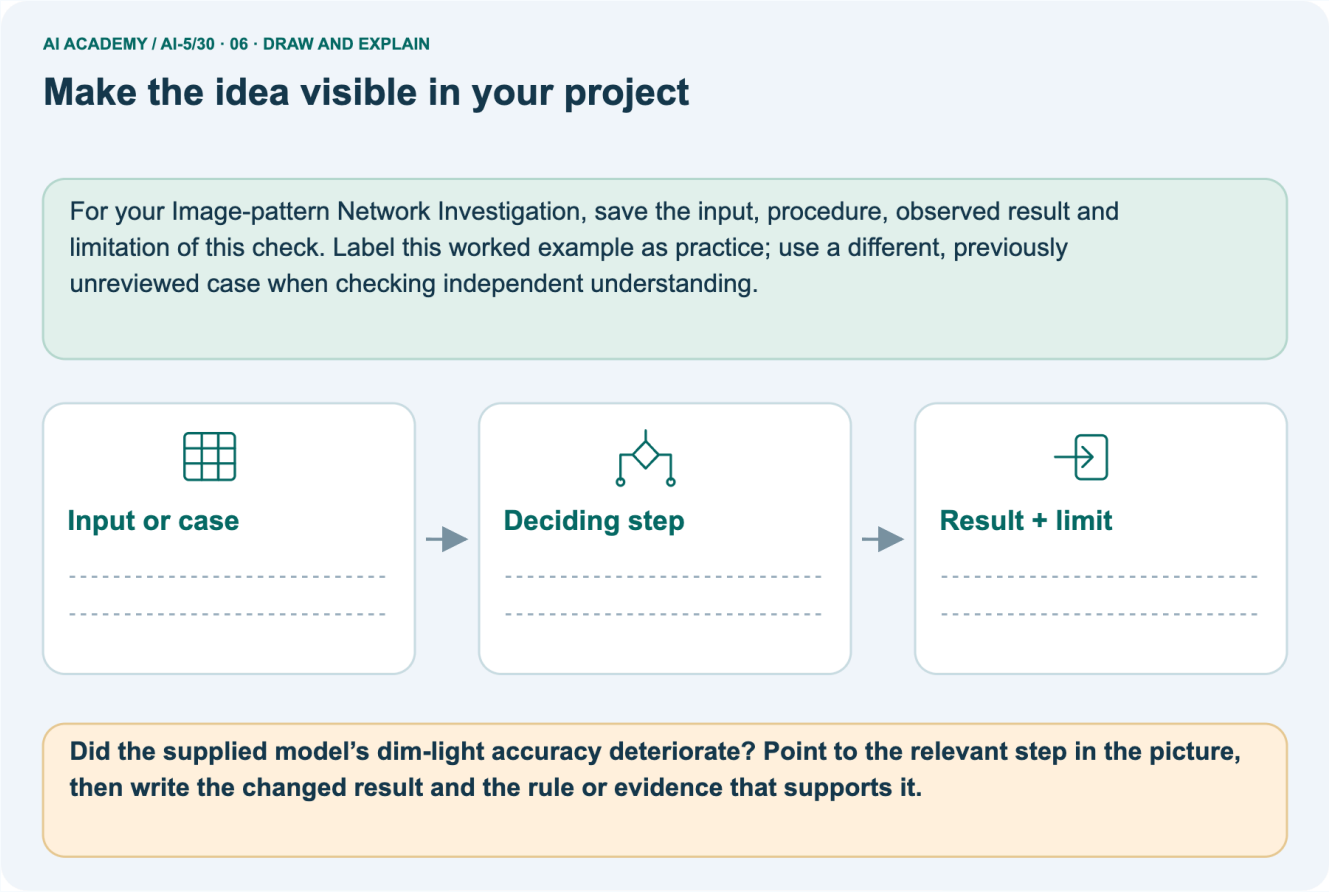
Limit: Real shift can also change labels, sensors and within-slice behaviour; this toy decomposition isolates only mixture change.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Real shift can also change labels, sensors and within-slice behaviour; this toy decomposition isolates only mixture change.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Did the supplied model's dim-light accuracy deteriorate? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No, it remains 50%. The pooled drop from 74% to56% reflects a larger proportion of dim cases, not changed within-slice rates. Changed-case check: Pooled accuracy returns to74%.

The mixture weights change, while each slice’s conditional performance stays fixed.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Dataset shift in neural models: the case

02. Follow the deciding steps

03. Fixed slice rates, changed pooled accuracy

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.